

AP CALCULUS AB – UNIT 3

Differentiation: Composite, Implicit, and Inverse Functions

Comprehensive Coloured Review



Name: _____ Class: _____ Date: _____

Review Scope and Learning Objectives

- Recognize composite structure and apply the Chain Rule in symbolic, tabular, and graphical settings.
- Differentiate implicit relations and determine tangent, normal, and higher-derivative information.
- Apply the derivative-of-an-inverse theorem from formulas, tables, and graphs.
- Differentiate inverse trigonometric functions and combine all derivative rules strategically.

Website Lesson Map

Lesson	Website title	Review target
3.1	The Chain Rule	Select and combine advanced differentiation procedures.
3.2	Implicit Differentiation	Select and combine advanced differentiation procedures.
3.3	Inverse Function and Inverse Trigonometric Derivatives	Select and combine advanced differentiation procedures.

Key Knowledge, Formulas, and Theorems

- Chain Rule: $\frac{d}{dx} f(g(x)) = f'(g(x))g'(x)$.
- Implicit differentiation treats y as $y(x)$, so $\frac{d}{dx}(y^n) = ny^{n-1}y'$.
- Inverse derivative: $(f^{-1})'(a) = \frac{1}{f'(f^{-1}(a))}$.
- $\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$, $\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$.
- Mark every layer before differentiating; multiply by the derivative of each inner layer.

Strategy and Precision Checklist

1. Identify the representation and the exact quantity requested before calculating.
2. Write the relevant definition, theorem, or derivative rule before substitution.
3. Preserve exact values in non-calculator work; use radian mode and appropriate rounding in calculator work.
4. For contextual conclusions, include the input, sign, units, and meaning.
5. Justifications must state why a theorem or procedure applies, not merely name it.

Solved Mixed Examples

Solved Example: Example 1 – 3.1

Differentiate e^{5x^3} .

Answer and Reasoning

Result: $15x^2 e^{5x^3}$. For e^u , keep the exponential and multiply by u' .

Solved Example: Example 2 – 3.1

Differentiate $\ln(x^2 + 1)$.

Answer and Reasoning

Result: $\frac{2x}{x^2+1}$. The derivative of $\ln u$ is u'/u .

Solved Example: Example 3 – 3.2

For the implicit curve $x^2 + xy + y^2 = K$, which formula for dy/dx is correct?

Answer and Reasoning

Result: $-\frac{2x+y}{x+2y}$. Differentiate xy using the product rule: $y + xy'$.

Solved Example: Example 4 – 3.2

For $x^2 + y^2 = 13$, find the slope at $(2, 3)$.

Answer and Reasoning

Result: $-\frac{2}{3}$. Implicit differentiation gives $2x + 2yy' = 0$, so $y' = -x/y$.

Leveled Mixed Practice

Foundation

Questions in this band integrate the website lessons at an increasingly demanding level.

1. Differentiate $f(x) = (3x - 1)^3$.

2. Find $\frac{d}{dx}[\sin(5x^2)]$.

3. Differentiate e^{5x^3} .

4. Differentiate $\ln(x^2 + 1)$.

Core

Questions in this band integrate the website lessons at an increasingly demanding level.

5. For the implicit curve $x^2 + xy + y^2 = K$, which formula for dy/dx is correct?

6. For $x^2 + y^2 = 13$, find the slope at $(2, 3)$.

7. If $f(1) = 4$ and $f'(1) = 2$, find $(f^{-1})'(4)$.

8. Differentiate $\arcsin(4x)$.

Advanced

Questions in this band integrate the website lessons at an increasingly demanding level.

9. Differentiate $\arctan(x^2)$.

10. Differentiate $x(x^2 + 2)^4$.

11. Differentiate $f(x) = (4x - 1)^3$.

12. Find $\frac{d}{dx}[\sin(3x^2)]$.

Difficult

Questions in this band integrate the website lessons at an increasingly demanding level.

13. Differentiate e^{4x^3} .

14. Differentiate $\ln(x^2 + 3)$.

15. For $x^2 + xy + y^2 = K$, estimate dy/dx at $(1.01, 1.11)$.

16. Use a calculator to evaluate the derivative of $\sin(3x)$ at $x = -0.46$.

Challenge

Questions in this band integrate the website lessons at an increasingly demanding level.

17. Find $\frac{d}{dx}(\arcsin x)$ at $x = -0.03$ to three decimals.

18. A numerical table gives $f(2.8) = 6.5$ and $f'(2.8) = 3.2$. Estimate $(f^{-1})'(6.5)$.

19. For $x^2 + xy + y^2 = K$, estimate dy/dx at $(1.59, 0.67)$.

20. Use a calculator to evaluate the derivative of $\sin(2x)$ at $x = 0.84$.

AP-Style Mixed Checkpoint

1. Differentiate $f(x) = (6x - 1)^4$.

(A) $24(6x - 1)^3$

(B) $4(6x - 1)^3$

(C) $24(6x - 1)^4$

(D) $6(6x - 1)^2$

2. Find $\frac{d}{dx}[\sin(4x^2)]$.

(A) $4 \cos(4x^2)$

(B) $\cos(4x^2)$

(C) $8x \cos(4x^2)$

(D) $8x \sin(4x^2)$

3. Differentiate e^{3x^3} .

(A) $9x^2 e^{3x^3}$

(B) e^{3x^3}

(C) $9x^2$

(D) $9x^2 e^{3x}$

4. Differentiate $\ln(x^2 + 3)$.

(A) $\frac{2x}{x^2+3}$

(B) $2x \log(x^2 + 3)$

(C) $\frac{1}{x^2+3}$

(D) $\frac{2}{x^2+3}$

5. For the implicit curve $x^2 + xy + y^2 = K$, which formula for dy/dx is correct?

(A) $-\frac{x+2y}{2x+y}$

(B) $-\frac{2x+y}{x+2y}$

(C) $-\frac{2x+1}{1+2y}$

(D) $\frac{2x+y}{x+2y}$

6. For $x^2 + y^2 = 41$, find the slope at $(4, 5)$.

(A) $\frac{5}{4}$

(B) $-\frac{4}{5}$

(C) $-\frac{5}{4}$

(D) $\frac{4}{5}$

7. If $f(4) = 2$ and $f'(4) = -2$, find $(f^{-1})'(2)$.

(A) -2

(B) $\frac{1}{2}$

(C) $\frac{1}{3}$

(D) $-\frac{1}{2}$

8. Differentiate $\arcsin(4x)$.

(A) $\frac{4}{1 + 16x^2}$

(B) $-\frac{4}{\sqrt{1 - 16x^2}}$

(C) $\frac{1}{\sqrt{1 - 16x^2}}$

(D) $\frac{4}{\sqrt{1 - 16x^2}}$

Free-Response Synthesis

FRQ1 – Nested Chain Rule

Let $f(x) = e^{(x^2+1)^3}$.

(a) Identify the three nested layers.

[2 points]

(b) Find $f'(x)$.

[3 points]

(c) Evaluate $f'(0)$.

[1 points]

FRQ2 – Implicit curve

The curve $x^2 + xy + y^2 = 7$ contains the point $(1, 2)$.

(a) Find dy/dx .

[3 points]

(b) Find the tangent slope at $(1, 2)$.

[1 points]

(c) Write the normal line at the point.

[2 points]

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